CROWDSURFING

Distributed Coordination for Emergent Happenings

Game Development Design Document

Version 1.0 — December 2025

An Organizational Game Design Framework Application

Integrating: Grassroots Economics • Holons • Trust Graph

Executive Summary

Crowdsurfing is a coordination protocol that transforms latent individual desires into emergent collective happenings. It addresses a fundamental coordination failure: people want to do things together but lack the infrastructure to discover aligned intentions, aggregate distributed resources, and activate commitment thresholds without requiring a hero organizer.

The system operates like Tinder for possibilities. Users declare what they want to make happen, what they're bringing, and what they need. Other users discover these possibility cards, opt in with their own resources, and when quorum is reached, coordination shifts from aggregation to execution via group messaging.

Post-event, participants express appreciation for contributions, feeding a trust graph that improves future coordination surfacing. Resources contributed accumulate as a commons library, making the network smarter over time.

Core Value Proposition

- **For individuals:** Transform 'I want to but I can't alone' into 'We did it together'
- **For communities:** Reduce coordination costs, surface latent capacity, build trust over time
- **For scenes:** Enable recurring coordination patterns without permanent organizational overhead

Technical Architecture Overview

Crowdsurfing integrates three existing open-source Ethereum-based projects, each handling distinct phases of the coordination loop:

1. **Grassroots Economics (Sarafu Network):** Handles possibility declarations, resource commitments, and time-boxing via demurrage
2. **Holons:** Activates group coordination and handles appreciation-weighted distribution
3. **Trust Graph / Free Association:** Maintains relational data from appreciation signals

Organizational Game Design Framework

Crowdsurfing is analyzed through the Organizational Game Design (OGD) framework, which treats coordination systems as games with intentionally designed rules, goals, feedback loops, and artifacts.

META (Green) — The Container

Meta elements define the possibility space itself—the rules that shape behavior, the frame that makes meaning possible.

Design Brief

Core Mechanic: Time-boxed possibility spaces with demurrage (use it or lose it pressure)

Invitation Structure: "I want to..." + time/space + requirements threshold

Fundamental Insight: This is a Design Brief protocol in software. Users design possibilities that recruit their own resources.

INTERACTION (Blue) — Formations & Figures

Interaction elements define the human patterns—how players move, what roles they take, the rhythm of coordination.

Roles

Role	Function	System Behavior
Possibility Declarer	Seeds coordination with desire + resources	Creates card, sets thresholds, commits first
Resource Contributor	Adds capacity to possibility	Commits resources, moves threshold bar
Need Spotter	Identifies gaps in coordination	Adds needs, inviting specific contributions
Appreciator	Signals trust post-event	Feeds trust graph, affects surfacing

Patterns

- **Tinder-style Card Surfing:** Low-friction discovery, binary interest signal, progressive detail reveal
- **Opt-in Escalation:** Interest → Commitment → Contribution (each stage reveals more, asks more)
- **Quorum Activation:** Group coordination only spawns when threshold met
- **Post-Event Appreciation Rounds:** Structured recognition flow after coordination succeeds

QUESTIONS (Red) — Directions & Development

Question elements define the longing—what we're reaching toward, the tension between here and there.

Invitation Sequence (Stringing)

1. **"What do you want to make happen?"** (Possibility)
2. **"What are you bringing?"** (Resources)
3. **"What would make this better?"** (Needs)
4. **"Who helped make this real?"** (Appreciation)

ARTIFACTS (Purple) — Maps & Marks

Artifact elements are what persists—the memory of coordination, the evidence that we were here.

Maps Created

- **Social Graph of Trust:** Who appreciated whom, weighted by frequency and reciprocity
- **Resource Library:** What capacities exist in the network (speakers, venues, skills, tools)
- **Coordination History:** What possibilities became real, what patterns repeat

Functional Specification

User Flows

Flow 1: Declare a Possibility

1. User taps 'Create Possibility'
2. Enters: "I want to... [grill in the park next weekend]"
3. Sets time window (start/end datetime)
4. Sets location (address, map pin, or 'TBD')

5. Sets participant cap (optional)
6. Declares initial resources: "I'm bringing... [a grill and a Frisbee]"
7. Declares needs: "It would be great if... [someone brought speakers, chicken, cups]"
8. Sets requirement thresholds (required vs. nice-to-have)
9. Publishes possibility card

Flow 2: Discover and Join

1. User opens app, sees stack of possibility cards
2. Cards sorted by: trust proximity, time urgency, resource match
3. Card shows: description, time/place, needs:resources progress bar, countdown
4. Swipe left: not interested (logged but not visible)
5. Swipe right: interested → expand card to full details
6. Full card shows: all needs, committed resources, participants
7. User commits: "I'm coming" + optional "I'm bringing... [X]"

Flow 3: Quorum Activation

1. System monitors threshold conditions
2. When all required needs met + minimum participants reached:
3. Holons creates group chat with all committed participants
4. Notification sent: "[Possibility] is happening!"
5. Chat pre-populated with: participant list, resource commitments, logistics

Flow 4: Post-Event Appreciation

1. After event end time, system prompts: "How did it go?"
2. User sees list of participants
3. For each participant, can express appreciation (tap or note)
4. Can highlight specific contributions: "[Person] brought [resource] and it was clutch"

5. Appreciation signals feed trust graph

Data Model

Core Entities

Possibility

Field	Type	Description
id	UUID	Unique identifier
creator_id	FK → User	Who declared this possibility
description	String (280 char)	"I want to..." text
time_start	DateTime	Event start time
time_end	DateTime	Event end time
deadline	DateTime	Coordination deadline (demurrage)
location	GeoJSON / String	Where it happens
min_participants	Integer	Required for quorum
status	Enum	draft|active|quorum|completed|expired
pool_address	ETH Address	Grassroots Economics pool contract

Need

Field	Type	Description
id	UUID	Unique identifier
possibility_id	FK → Possibility	Which possibility needs this
description	String	"It would be great if..." text
is_required	Boolean	Required for quorum vs nice-to-have
category	String (tag)	equipment|venue|skill|person|consumable
fulfilled_by	FK → Commitment	Which commitment satisfies this

Additional Entities (summarized)

- **User:** id, wallet_address, display_name, trust_score, created_at
- **Commitment:** id, possibility_id, user_id, resource_description, committed_at, status
- **Appreciation:** id, from_user_id, to_user_id, possibility_id, note, created_at
- **Resource:** id, owner_id, description, category, availability_status, times_contributed
- **TrustEdge:** from_user_id, to_user_id, weight, last_updated

Technical Architecture

Integration Architecture

Crowdsurfing orchestrates three existing open-source Ethereum-based systems. Rather than building from scratch, we compose existing battle-tested infrastructure.

Grassroots Economics (Sarafu Network)

Role: Commitment Pooling & Time-Boxing

License: AGPL 3.0

Repository: github.com/grassrootseconomics

- Possibility declarations create commitment pools
- Resource commitments represented as Community Asset Vouchers (CAVs)
- Time-boxing implemented via demurrage mechanisms
- eth-custodial for key management (non-crypto-native users)

Holons

Role: Group Coordination & Appreciation-Weighted Distribution

Repository: github.com/liminalvillage/holons

- Quorum activation spawns Holon for participant group
- Stackable payment splitter for any pooled resources
- Appreciation signals weight distribution ratios

Trust Graph / Free Association

Role: Relational Data & Surfacing Weights

Repository: github.com/interplaynetary/free-association (playnet.lol)

- Mutual recognition from appreciations
- Trust-weighted possibility surfacing
- Permission boundaries for resource sharing

Technology Stack

Layer	Technology	Rationale
Mobile Client	React Native / Expo	Cross-platform, rapid iteration
Web Client	Next.js / React	SEO, shared component library
API Layer	Node.js / TypeScript	Matches GE stack, type safety
Database	PostgreSQL + Redis	Relational + real-time state
Blockchain	Celo (EVM-compatible)	GE ecosystem, low gas fees
Real-time Comms	Matrix Protocol	Decentralized, E2E encrypted
Feature Phone	USSD via sarafu-vise	Non-smartphone accessibility

Project Planning

Team Composition

Role	Level	Primary Focus
Developer	Senior	Architecture, blockchain integration, code review
Developer	Mid-level	Feature development, API layer, testing
Developer	Junior	UI components, documentation, QA support

Scrum Master	Experienced	Sprint planning, blockers, stakeholder comms
Designer	Product/UX	User flows, wireframes, visual design
Product Manager	-	Roadmap, prioritization, user research
Analyst	-	Metrics, user behavior, integration specs
Brand Strategist	-	Positioning, voice, community narrative

Complexity Assessment

Overall Complexity: HIGH

This is not a simple CRUD app. Crowdsurfing integrates three distinct blockchain-based systems, requires real-time coordination, and depends on network effects for value.

- **Multi-system integration:** Three separate open-source projects with their own paradigms
- **Blockchain UX abstraction:** Users shouldn't know they're interacting with Ethereum
- **Real-time state sync:** Possibility status must sync across clients instantly
- **Cold start problem:** Network effects required for value

Development Roadmap

Total estimated duration: 9-12 months to production-ready MVP

Phase 0: Foundation (Weeks 1-4)

Goal: Establish project infrastructure, team alignment, integration feasibility

- Development environment setup (monorepo, CI/CD, staging)
- Integration feasibility spikes with GE, Holons, Trust Graph teams
- Brand identity and design system foundations
- User research: 10+ interviews with target community members

Phase 1: Core Loop (Weeks 5-12)

Goal: Build minimal coordination loop: declare → discover → join → coordinate

- Mobile app: Discovery feed, possibility detail, create possibility
- API: CRUD for possibilities, needs, commitments
- Basic quorum detection (no blockchain yet—server-side)
- Simple group chat via Matrix integration

Phase 2: Blockchain Integration (Weeks 13-22)

Goal: Connect to Grassroots Economics for commitment pooling and time-boxing

- Custodial wallet creation (non-crypto UX)
- Commitment pool creation per possibility (Sarafu integration)
- Demurrage mechanism for time-boxing
- Holon creation on quorum activation

Phase 3: Trust Layer (Weeks 23-30)

Goal: Implement appreciation flow and trust-weighted surfacing

- Post-event appreciation flow (mobile + web)
- Trust Graph integration (Free Association / playnet.lol)
- Trust-weighted possibility surfacing algorithm
- Resource library (commons of contributed capacities)

Phase 4: Polish & Launch (Weeks 31-40)

Goal: Production readiness, community seeding, public launch

- Security audit (smart contract + API)
- Performance optimization (load testing, caching)
- Community seeding: 3-5 pilot communities
- App store submissions (iOS + Android)

Timeline Summary

Phase	Duration	Weeks	Sprints	Exit Criteria
0: Foundation	1 month	1-4	2	Go/no-go decision
1: Core Loop	2 months	5-12	4	Internal dogfooding
2: Blockchain	2.5 months	13-22	5	Testnet launch
3: Trust Layer	2 months	23-30	4	Closed beta
4: Polish & Launch	2.5 months	31-40	5	Public launch

Risks & Mitigations

Technical Risks

- **Integration dependency:** Three external projects may have API changes. Mitigation: Early engagement with maintainers, abstraction layers.
- **Blockchain UX friction:** Gas fees, wallet management confuse users. Mitigation: Custodial wallet, meta-transactions, optimistic UI.

- **Smart contract security:** Bugs could lead to fund loss. Mitigation: Use battle-tested GE contracts, third-party audit.

Product Risks

- **Cold start / network effects:** Empty feed provides no value. Mitigation: Seed with pilot communities, city-by-city rollout.
- **Coordination failure rates:** Too many expired possibilities. Mitigation: Smart defaults for thresholds, gentle nudges.

Success Metrics

North Star Metric

Successful Coordinations per Week: Possibilities that reach quorum and result in post-event appreciation

Funnel Metrics

Stage	Metric	Target (6mo)
Awareness	App installs	10,000
Activation	First commitment made	40% of installs
Engagement	Weekly active users	2,000
Retention	Month-3 retention	25%
Coordination Success	Quorum rate	>50%

Network Health	Appreciation ratio	>0.8

Appendices

A. Glossary

- **Possibility:** A declared coordination opportunity ("I want to...")
- **Quorum:** The threshold of participants and resources required to activate coordination
- **Demurrage:** Time-decay mechanism that creates urgency (use it or lose it)
- **CAV (Community Asset Voucher):** Grassroots Economics token representing commitment to goods/services
- **Holon:** Self-organizing unit that can nest within larger structures; Ethereum-based payment splitter
- **Trust Graph:** Network of mutual recognition relationships derived from appreciation signals
- **Mutual Recognition:** Bidirectional acknowledgment of contribution (minimum of both parties' appreciation)

B. Related Projects & Documentation

- Grassroots Economics: docs.grassecon.org, github.com/grassrootseconomics
- Holons: github.com/liminalvillage/holons, holons.io
- Free Association / Trust Graph: github.com/interplaynetary/free-association, playnet.lol
- Sarafu Network: sarafu.network
- Matrix Protocol: matrix.org

C. OGD Framework Reference

This document applies the Organizational Game Design framework:

- **Meta (Green):** Design Brief, Select & Deploy — the container and rules
- **Interaction (Blue):** Roles, Patterns — the human movements
- **Questions (Red):** Compass, Stringing — the direction and longing
- **Artifacts (Purple):** Maps, Marks — what persists after

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Prepared using the Organizational Game Design framework

For development team use